Khulna University of Engineering & Technology
B. Sc. Engineering 1st Year 2nd Term Regular Examination, 2023
Department of Materials Science and Engineering
Hum 1227
Technical English

Time: 3 hours | Full Marks: 210

N.B.: (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figures in the right margin indicate full marks.

Section A

Q1. (a) Make sentence with the following structures using the words given in brackets. (14)

(i) Subj. + Intransitive Verb + Adv. of time. (walk as verb)
(ii) Subj. + Linking Verb + Adj. Complement. (seem as verb)
(iii) Why + Subj. + Verb + Adv. of place + Verb + Adj. Complement. (come and is as verb)
(iv) Subj., + Relative Pronoun + Verb + Adv. of place + Verb + Adj. Complement. (work and is as verb)
(v) Since + Subj. + Verb + Adj. Complement, Subj. + Verb + Obj. (is and like as verb)
(vi) Subj. + Verb + Adv. of manner + so that + Subj. + Verb + Adv. of place. (live and progress as verb)
(vii) Neither + Subj. + nor + Subj. + Verb + Adv. of place. (attend as verb)

(b) Change the following words as asked in brackets and use the changed forms in sentence. (12)

Calamity (into adj.), Deep (into noun), Acclimatisation (into verb), Abandonment (into verb), Use (into noun), Class (into adj.)

(c) Make a new word with the following prefixes and suffixes and use the new words in sentence. (09)

Ab–, Mega–, Im–, –ian, –tude, –worthy

Q2. (a) Transform the following sentences as asked in brackets. (14)

(i) Everyone will realise that I am right. (Negative)
(ii) It does not matter if I go. (Interrogative)
(iii) Mim writes rapidly that she can complete answer in time. (Simple)
(iv) At eighteen Saju would walk a long path. (Complex)
(v) Besides living peacefully he enjoys life. (Compound)
(vi) Habib looks as smart as other students. (Comparative)
(vii) What cannot be cured must be endured. (Active)

(b) Make use of the following words in sentence as asked in brackets. (12)

Cheer (as verb), Above (as adverb), Till (as preposition), Till (as conjunction), School (as adjective), Go (as noun)

(c) Write one antonym and one synonym for each of the words given below and use the antonyms and synonyms in sentence. (09)

Delicious, Intelligence, Intimate

Q3. (a) Make wh question with the underlined words of the following sentences. (14)

(i) Helen met us yesterday.
(ii) The meeting lasted an hour and a half.
(iii) The train is running at a speed of 90 Kms per hour.
(iv) He is about average height.
(v) He completes the duty with a joyous heart.
(vi) Mim comes here to see the situation.
(vii) I gave a doll to a little girl.

(b) Complete the following sentences with clauses as asked in brackets. (12)

(i) —— touches our feeling. (noun clause)
(ii) It is noteworthy ——. (noun clause)
(iii) Rakib, ——, follows his duties sincerely. (adj. clause)
(iv) I will stay ——. (adv. clause of time)
(v) ——, we can't help you. (adv. clause of reason)
(vi) ——, he is not gentle. (adv. clause of concession)

(c) Express the following notions and functions in sentence. (09)

(i) Tension, (ii) Liable, (iii) Honesty, (iv) Thanks, (v) Excuse, (vi) Farewell

Q4. (a) Correct the following sentences. (14)

(i) The Oxygen is essential for humans.
(ii) Bangladesh Biman will land within a short time.
(iii) Quote the lines from heart.
(iv) He need not to go there.
(v) It is raining since morning.
(vi) He is true to his words.
(vii) The chair's legs are broken.

(b) Make use of the following modal auxiliary verbs in sentence as asked in brackets. (12)

(i) Can. (To express offer to somebody else)
(ii) Could. (To express a past ability)
(iii) Would. (To express a polite request)
(iv) Used to. (To express a past regular habit)
(v) Dare. (To express indulgence)
(vi) Must + have + past participle of verb. (To express a logical deduction)

(c) Make sentence with the following idioms and phrases. (09)

Cut somebody some slack, Cutting corners, To get bent out of shape, Pull yourself together, Elbow grease, Forty winks

Section B

Q5. (a) Read the passage and answer the questions that follow. (20)

Silence is unnatural to man. He begins life with a cry and ends it in stillness. In the interval he does all he can to make a noise in the world, and there are a few things of which he stands in more fear than of the absence of noise. Even his conversation is in great measure a desperate attempt to prevent a dreadful silence. If he is introduced to a fellow mortal and a number of pauses occur in the conversation, he regards himself as a failure, a worthless person, and is full of envy of the emptiest headed chatter-box. He knows that ninety nine percent of human conversation means no more than the buzzing of a fly, but he longs to join in the buzz, and to prove that he is a man and not a wax-work figure. The object of conversation is not, for the most part, to communicate ideas; it is to keep up the buzzing sound. There are, it must be admitted, different qualities of buzz. Most buzzing fortunately is pleasing to the ear, some of it is agreeable to the mind. Very few human beings join in the conversation in the hope of learning anything new. Some of them are content if they are merely allowed to go on making a noise into other people's ears, though they have nothing to tell them except that they have seen two or three new plays or that they had bad food in a Swiss hotel.

Questions:
(i) How does the author prove that silence is unnatural?
(ii) When does a man consider himself a worthless person?
(iii) How does the author characterise human conversation?
(iv) What, according to the author, is the object of conversation?

(b) Make a précis of the passage written above (Q5(a)) with a suitable title. (15)

Q6. (a) Write a cause and effect paragraph on vanishing social values. (15)
(b) Amplify the idea contained in the following statement. (20)
Experience is the best teacher.

Q7. (a) Suppose you are the Production Manager of a company. An employee has recently been promoted to the position of a supervisor. Write a memo to all employees of the company informing his promotion. (15)
(b) Write a report on economic condition of your surroundings to publish in a daily newspaper. (20)

Q8. Write a free composition on any one of the followings: (35)
(i) The dowry system and women in Bangladesh
(ii) Hardworking: A key to happiness in life

Khulna University of Engineering & Technology

B. Sc. Engineering 1st Year 2nd Term Regular Examination, 2023

Department of Materials Science and Engineering

Math 1227

Differential Equations and Geometry

Time: 3 hours Full Marks: 210

N.B.: (i) Answer **ANY THREE** questions from each section in separate scripts.

(ii) Figures in the right margin indicate full marks.

Section A

Q1. (a) Define order and degree of a differential equation. Also, define trajectory. (06)

(b) Obtain the differential equation associated with the primitive $y = Ae^{2x} + Be^{x} + C$, where A, B and C are arbitrary constants. (10)

(c) Solve the initial value problem $x\sin y dx + (x^2+1)\cos y dy = 0, y(1) = \frac{\pi}{2}$. (10)

(d) Find the orthogonal trajectories of the family of coaxial circles $x^2 + y^2 + 2gx + c = 0$, where g is an arbitrary constant and c is a fixed constant. (09)

Q2. (a) Find the general solution of the differential equation $(x + 2y - 3)dx - (2x + y - 3)dy = 0$. (13)

(b) Solve $(y - xy^2)dx - (x + x^2y)dy = 0$. (11)

(c) Solve $xdy - \{y + xy^3(1 + \ln x)\}dx = 0$. (11)

Q3. (a) Solve $(D^2 - 6D + 9)y = \frac{e^{3x}}{x^2}$, where $D = \frac{d}{dx}$. (12)

(b) Find the general solution of the differential equation $(D^2 - 2D + 1)y = x\sin x$. (11)

(c) Solve $(D^2 - 4D + 3)y = e^{3x}x\cos 5x$, where $D = \frac{d}{dx}$. (12)

Q4. (a) Define Laplace transform. (02)

(b) Find the Laplace transform of $t^3 - 9e^{5t} + 14\sin 4t - 25\cos 3t$. (09)

(c) Find the Laplace transform of $F(t)$ where $F(t) = \begin{cases} \cos\left(t - \frac{2\pi}{3}\right), & t > \frac{2\pi}{3} \\ 0, & t < \frac{2\pi}{3} \end{cases}$ (12)

(d) Find the Laplace transform of the periodic function $F(t)$ with period 2π such that $F(t) = \sin t$ for $0 < t < \pi$ and $F(t) = 0$ for $\pi < t < 2\pi$. (12)

Section B

Q5. (a) Find the coordinates $\left(3, \frac{5\pi}{4}, -3\right)$ in Cartesian and spherical polar coordinates. (10)

(b) Define direction ratio and direction cosines of a line. Find the angle between the lines whose direction cosines are given by the relations $l - 2m + 3n = 0$ and $l^2 - 4m^2 - 9n^2 = 0$. (13)

(c) A line makes angle α, β, γ and δ with the four diagonals of a cube, show that $\cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta = \frac{4}{3}$. (12)

Q6. (a) Find the equation of the plane through the points (2,2,1) and (9,3,6) and perpendicular to the plane $2x + 6y + 6z = 9$. (13)

(b) A variable plane is at a constant distance P from the origin, meets the axes at A, B, C planes through A, B, C parallel to the coordinate planes meets at θ. Find the locus of θ. (10)

(c) Define Skew lines. Find the magnitude and equation of the line of shortest distance between the lines $\frac{1}{3}(x-3) = -(y-8) = (z-3)$ and $-\frac{1}{3}(x+3) = \frac{1}{2}(y+7) = \frac{1}{4}(z-6)$. (12)

Q7. (a) Test whether the lines $\frac{x-1}{2} = \frac{y+2}{-3} = \frac{z-4}{5}$ and $2x+3y+4z-4 = 0 = x+y-z+7$ are coplanar or not. If possible find the equation of the plane containing them. (12)

(b) Find the center and radius of the circle in which the sphere $x^2+y^2+z^2+2y+4z-11=0$ is cut by the plane $x+2y+2z+15=0$. (13)

(c) Find the angle between the line $2x+4y-2z+3=0=4x-2y+2z+5$ and the plane $5x-4y+3z-5=0$. (10)

Q8. (a) Find the equations of line perpendicular to both the line $(x-1) = \frac{1}{2}(y-1) = \frac{1}{3}(z+2)$ and $\frac{1}{2}(x+2) = -(y-5) = \frac{1}{2}(x+3)$ and passing through their intersection. (11)

(b) Find the equation of the sphere for which the circle $x^2+y^2+z^2-2x+4y-6z-9=0,\ 2x+3y-4z+5=0$ is a great circle. (12)

(c) Find the equation of the sphere which touches $3x+y-4z-12=0$ at $(1,1,-2)$ and is orthogonal to the sphere $x^2+y^2+z^2-4x+6y-8z-7=0$. (12)

Khulna University of Engineering & Technology
B. Sc. Engineering 1st Year 2nd Term Regular Examination, 2023
Department of Materials Science and Engineering
Ph 1227
Magnetism and Nuclear Physics

Time: 3 hours | Full Marks: 210

N.B.: (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figures in the right margin indicate full marks.

Section A

Q1. (a) Explain the term magnetic dipole and electric dipole moment. (08)

(b) Find the value of magnetic field B due to a current loop of radius R carrying a current i at a distance x on the axis from the centre of the loop. (15)

(c) A coil having an area of 2.52 cm^2 and 250 turns carries a current of 85 μA. (12)
(i) What is the magnetic dipole moment of the coil.
(ii) The magnetic dipole moment of the coil is linked with an external magnetic field whose strength is 0.85 T. How much work would be done by the external agent to rotate the coil through 180°?

Q2. (a) Define magnetic circuit. Make a comparison between electric circuit and magnetic circuit. (10)

(b) State Biot-Savart law. Applying this law find the value of 'B' due to a long straight wire carrying a current. (15)

(c) The perpendicular component of the external magnetic field through a 10-turn coil of radius 50 mm increases from 0 to 18 T in 3 sec. If the resistance of the coil is 2 Ω, what is the magnitude of the induced current? (10)

Q3. (a) Explain the cause of Hysteresis phenomenon in ferromagnetic materials. What does the area of the Hysteresis loop signify? (10)

(b) Derive an expression for diamagnetic susceptibility according to Langevin's classical theory. (15)

(c) An iron rod of density 7.7 gm/c.c. and specific heat 0.1 cal/gm/°C is subjected to magnetization of 50 cycles per second. If the area enclosed by $I_m - H$ curve for the specimen is 5×10^4 erg, find the rise in temperature per minute ($J = 4.2\times10^7$ erg/cal). (10)

Q4. (a) Define (i) Dielectric constant (ii) Polarizability (iii) Piezoelectricity and (iv) Pyroelectricity. (12)

(b) Derive the Clausius-Mossotti relation expressing the relationship between dielectric constant and atomic polarizability. (13)

(c) Find the total polarizability of CO_2, if its susceptibility is 0.985×10^{-3}. The density of carbon dioxide is 1.977 Kg/m^3. (10)

Section B

Q5. (a) Explain the general behaviour of material under stress using a stress-strain diagram. (10)

(b) If Y, K and n represent Young's modulus, Bulk modulus and modulus of rigidity respectively, then derive the following relation connecting the three elastic (15) constants: $\frac{3}{\eta} + \frac{1}{K} = \frac{9}{Y}$.

(c) The Young's modulus of a metal is 2×10^{11} N/m^2 and its breaking stress is 1.078×10^{9} N/m^2. Calculate the maximum amount of energy per unit volume which can be stored in the metal when stretched. (10)

Q6. (a) Define (i) Binding energy (ii) Mass defect (iii) Packing fraction and (iv) Nuclear magnetic moment. (12)

(b) Draw the binding energy per nucleon versus mass number curve. Comment about nuclear fission and fusion reactions from the curve. (13)

(c) Calculate the total binding energy of ${}_{1}^{2}D$ and ${}_{10}^{20}Ne$. Given $m_p = 1.007825$ amu, $m_n = 1.008665$ amu, $m_D = 2.0141$ amu and $m_{Ne} = 19.992439$ amu. (10)

Q7. (a) Explain the laws of radioactive decay. From the decay law obtain an expression for half-life of a radioactive sample. (10)

(b) Develop a mathematical formula for successive decay of radioactive substance and discuss permanent equilibrium using this formula. (15)

(c) 1 gram of Radium is reduced to 2.2 milligram in 5 years by α-decay. Calculate the half-life and mean life of Radium. (10)

Q8. (a) Determine that the couple required per unit twist in the case of a cylinder is $C = \frac{\pi n r^4}{2l}$, where symbols have their usual meaning. (10)

(b) Describe briefly the various components of a nuclear reactor. Explain the principle of its working. (15)

(c) Assuming the energy released per fission of U^{235} is 200 MeV. Determine the rate in Kg per year at which fission should occur in a nuclear reactor in order to operate at a power level of 1 watt. (10)

Khulna University of Engineering & Technology

B. Sc. Engineering 1st Year 2nd Term Regular Examination, 2023

Department of Materials Science and Engineering

Ch 1227

Organic Chemistry

Time: 3 hours | Full Marks: 210

N.B.: (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figures in the right margin indicate full marks.

Section A

Q1. (a) Sketch the molecular orbital diagram of benzene and delineate the feature of it. (08)

(b) Construe the mechanism of Friedel-Craft acylation reaction with activation energy diagram. (10)

(c) "Benzene does not give electrophilic addition reaction though benzene has three double bonds" Why? (09)

(d) Indicate why polyalkylation and branching are occurred in Friedel-Craft alkylation reaction. (08)

Q2. (a) Dictate the instrumentation of UV-Vis. spectroscopy with description of each part. (10)

(b) What type of electronic transitions occurs in organic molecule? (09)

(c) Absorption bands in UV-Vis. spectra are broad in nature – Justify the statement. (08)

(d) A conjugated diene absorbs at higher wavelength than diene where double bonds are isolated – Demonstrate. (08)

Q3. (a) Why is methanol a good solvent for UV but not for IR determination? (07)

(b) Rank the following based on increasing order of wave number for – C = O group. Given reasons. (10)

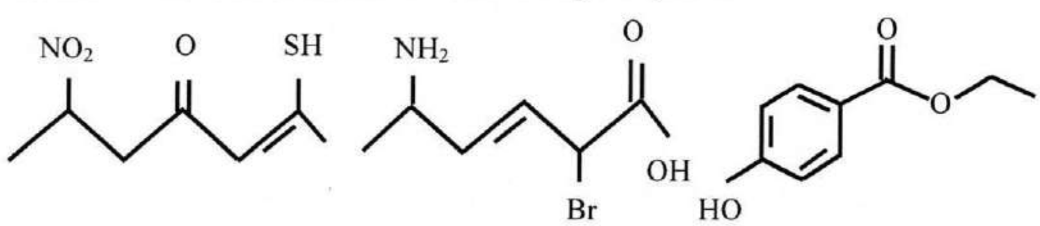
Tetrachloroacetone, Acetone, Dichloroacetone, Chloroacetone.

(c) Depict the effect of alkyl group on double bond (C = C) absorption frequency. (09)

(d) Define fingerprint region. Concisely illustrate different fingerprint region with example. (09)

Q4. (a) Neopentane is gas while n-pentane is liquid – Infer based on London dispersion forces. (08)

(b) Provide IUPAC nomenclature of the following compound. (09)



(c) Out of cis and trans isomers, which one has higher melting point and why? (08)

(d) Draw all possible isomers of C_4H_9OH. Comment on the boiling point of the isomers. (10)

Section B

Q5. (a) Classify polymers based on branching with examples. (05)

(b) Define with examples of the followings (i) Living polymer (ii) Monomeric unit (iii) Natural polymer (iv) Heterochain polymer. (12)

(c) "All polymers are macromolecules, but all macromolecules are not polymer" Why? (08)

(d) Synthesize the following polymers: (10)
(i) Conducting polymer (ii) Nylon-66.

Q6. (a) What is degree of polymerization? Discuss the mechanism of free radical addition polymerization reaction. (12)

(b) Differentiate between thermosetting polymer and thermoplastic polymer. (08)

(c) What is Stereospecific polymer? Give examples. (07)

(d) A polymer consists of 9 gram having molecular weight 30,000 and 5 gram having molecular weight 50,000. Calculate the number average molecular weight and weight average molecular weight. (08)

Q7. (a) What is the Stereochemical approach of S_N1 and S_N2 reactions? (12)

(b) Why do we get isopropyl benzene on treating benzene with 1–chloropropane instead of n–propyl benzene? Describe briefly. (12)

(c) Write down and explain the E1 mechanism. (11)

Q8. (a) Draw the structure of cellulose. What happens when cellulose is treated with acetic anhydride in the presence of acetic acid and small amount of concentrated sulphuric acid? (09)

(b) Describe two methods for the synthesize of amino acids. (10)

(c) What is isoelectric point of amino acids? Demonstrate with chemical equilibrium. (08)

(d) Shortly construe the denaturation process of proteins. (08)

Khulna University of Engineering & Technology

B. Sc. Engineering 1st Year 2nd Term Regular Examination, 2023

Department of Materials Science and Engineering

ME 1227

Engineering Mechanics

Time: 3 hours | Full Marks: 210

N.B.: (i) Answer **ANY THREE** questions from each section in separate scripts.

(ii) Figures in the right margin indicate full marks.

(iii) Assume any reasonable data if missing.

Section A

Q1. (a) The bodies A, weighing 50 N and B, weighing 75 N are connected by a cord and resting on smooth plane. What is the angle θ if the bodies are in equilibrium? Also, determine the reaction forces. (17)

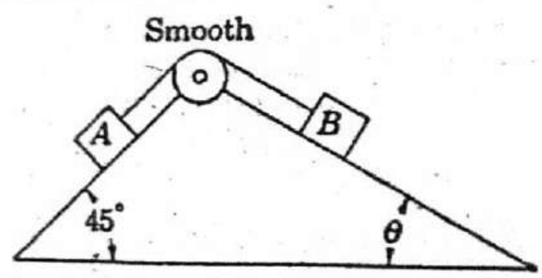


Fig. for Q1(a)

(b) A container of weight W is suspended from ring A. Cable BAC passes through the ring and is attached to fixed supports at B and C. Two forces $\vec{P} = P\underline{i}$ and $\vec{Q} = Q\underline{k}$ are applied to the ring to maintain the container in the position shown. Knowing that $W = 376$ N, determine P and Q. (18)

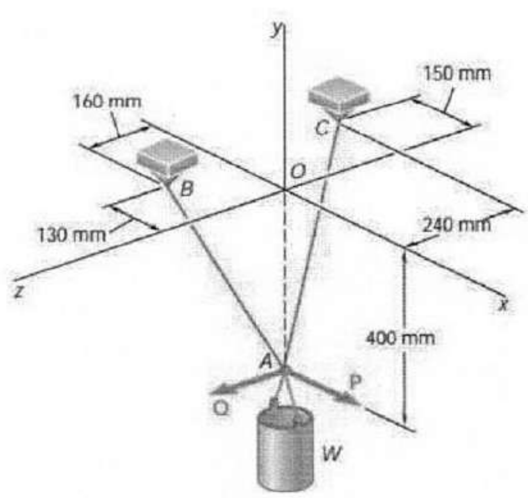


Fig. for Q1(b)

Q2. (a) The 6 m boom AB has a fixed end A (Fig. Q2(a)). A steel cable is stretched from the free end B of the boom to a point C located on the vertical wall. If the tension in the cable is 2.5 kN, determine the moment about A of the force exerted by the cable at B. (17)

(b) A 300 N force is applied at A as shown in Fig. Q2(b). Determine (a) the moment of the 300 N force about D, (b) the smallest force applied at B that creates the same moment about D. (18)

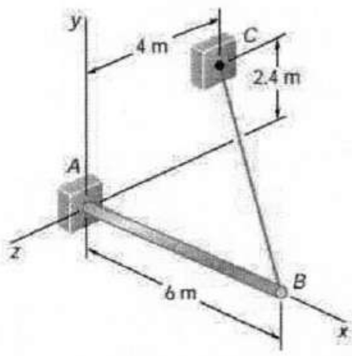


Fig. for Q2(a)

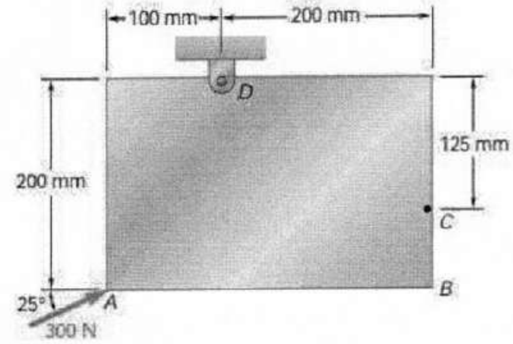


Fig. for Q2(b)

Q3. (a) Two links AB and DE are connected by a bell crank as shown. Knowing that the tension in link AB is 720 N, determine (a) the tension in link DE, (b) the reaction at C. (17)

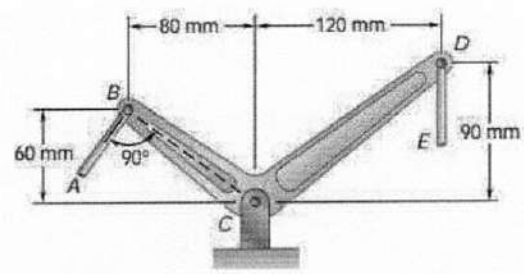


Fig. for Q3(a)

(b) Determine the reactions at A and B when – (i) $\alpha = 0°$ (ii) $\alpha = 90°$ (iii) $\alpha = 30°$. (18)

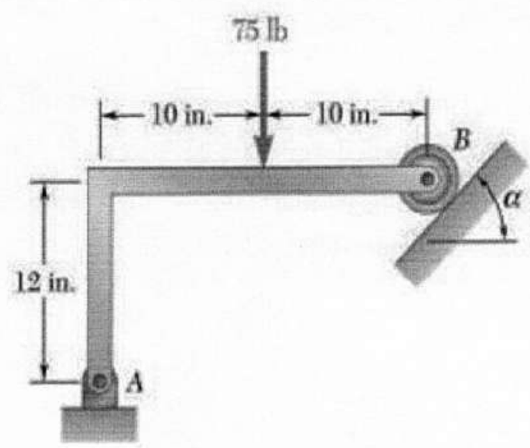


Fig. for Q3(b)

Q4. (a) A light bar AD is suspended from a cable BE and supports a 50 N block at C. The ends A and D of the bar are in contact with frictionless vertical walls. Determine the tension in cable BE and the reactions at A and D. (17)

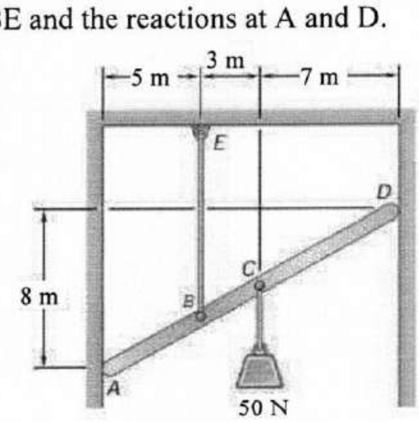


Fig. for Q4(a)

(b) A 2.4 m boom is held by a ball and socket joint at C and by two cables AD and AE. Determine the tension in each cable and the reaction at C. (18)

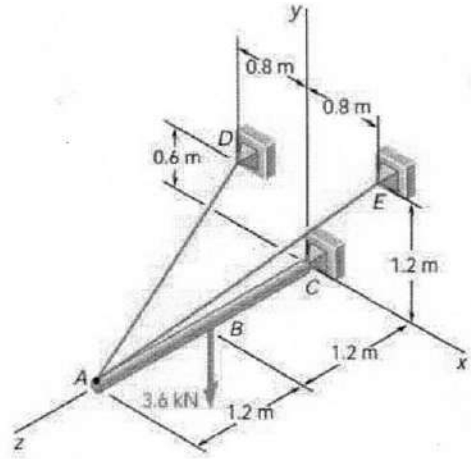


Fig. for Q4(b)

Section B

Q5. (a) Describe the principle of conservation of linear momentum for a particle. (05)

(b) Milk is poured into a glass of height 140 mm and inside diameter 66 mm. If the initial velocity of the milk is 1.2 m/s at an angle of 40° with the horizontal, determine the range of values of height h for which the milk will enter the glass. (14)

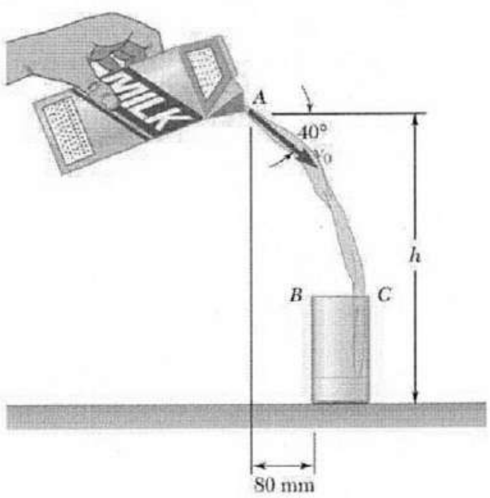


Fig. for Q5(b)

(c) Block A has a mass of 25 Kg and block B has a mass of 15 Kg. The coefficients of friction between all surfaces of contact are $\mu_s = 0.20$ and $\mu_k = 0.15$. Knowing $\theta = 30°$ and the magnitude of the force P applied to block A is 250 N, determine- (16)

(i) The acceleration of block A

(ii) The tension in the cord.

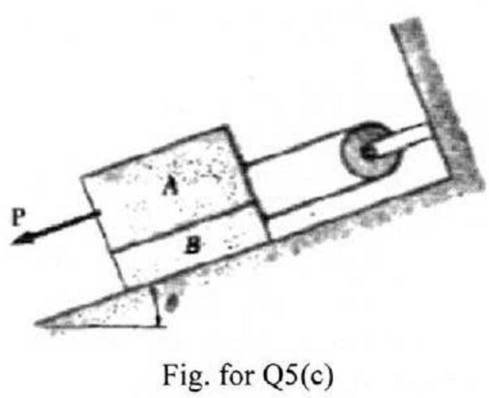


Fig. for Q5(c)

Q6. (a) A drum of 100 mm radius is attached to a disk of 200 mm radius. The disk and drum have a combined weight of 4.54 Kg and a combined radius of gyration of 150 mm. A cord is attached as shown in Fig. Q6(a) and pulled with a force P of magnitude 22 N. Knowing that the coefficients of static and kinetic friction are (17)

$\mu_s = 0.25$ and $\mu_k = 0.20$ respectively, determine the angular acceleration of the disk and the acceleration of G.

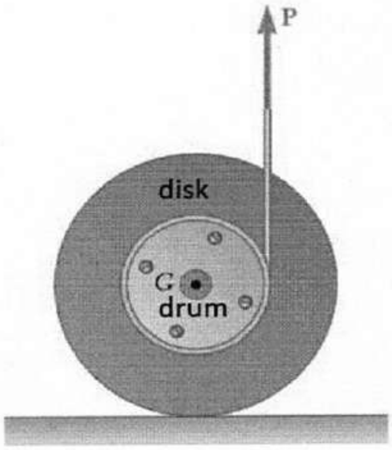


Fig. for Q6(a)

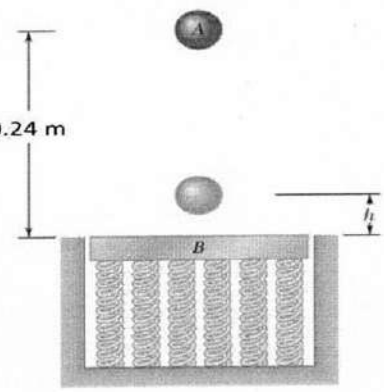


Fig. for Q6(b)

(b) A 0.59 Kg sphere A is dropped from a height of 0.57 m onto a 1.18 Kg plate B, which is supported by a set of springs and is initially at rest (Fig. Q6(b)). Knowing that the coefficient of restitution between the sphere and the plate is $e = 0.8$, determine the height reached by the sphere after rebound. (18)

Q7. (a) Small wheels have been attached to the ends of rod AB and roll freely along the surfaces shown. Knowing that wheel A moves to the left with a constant velocity of 1.5 m/s, determine – (i) angular velocity of the rod, (ii) the velocity of end B of the rod. (17)

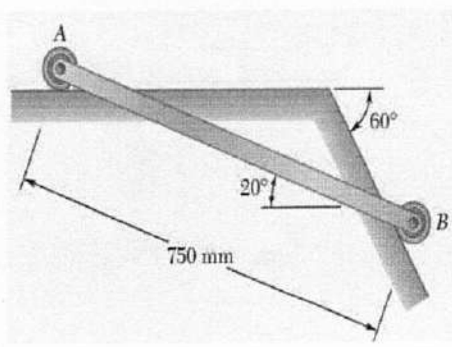


Fig. for Q7(a)

(b) Explain briefly about the turning moment diagram for a four stroke internal combustion engine with neat sketch. (09)

(c) A horizontal steam engine develops 300 kW at 90 rpm. The coefficient of fluctuation of energy as found from the turning moment diagram is to be 0.1 and the fluctuation of speed is to be kept within $\pm 0.5\%$ of the mean speed. Find the weight of the flywheel required, if the radius of gyration is 2 m. (09)

Q8. (a) How can you relate controlling force of a governor with the stability of a governor? What is meant by governor effort? (08)

(b) Explain the working principle of proell governor with neat sketch. (10)

(c) A porter governor has links 150 mm long and are attached to pivots at a radial distance of 30 mm from the vertical axis of the governor. The mass of each ball is 1.75 Kg and the mass of the sleeve is 25 Kg. The governor sleeve begins to rise at 300 rpm when the links are at 30° to the vertical. Assuming the friction force to be constant, find the maximum and minimum speed of rotation when the inclination of the links is 45° to the vertical. (17)